Project Details:-

1. Healthcare Smart Care Bot (Generative AI Project)

Team Size: 4 members
Project Duration: ~8 weeks

Project Summary:
The Healthcare Smart Care Bot is a **Generative AI–based assistant** designed to support users with healthcare-related queries such as symptom guidance, general medical information, and personalized responses using trusted data sources. The main objective was to create a reliable, context-aware AI system that avoids hallucinations and provides accurate information.

My Role & Contributions:

- Fine-tuned **Large Language Models (LLMs)** using Hugging Face Transformers for healthcare-related queries.
- Implemented **Retrieval Augmented Generation (RAG)** using LangChain to fetch verified medical information before generating responses.
- Performed data preprocessing and prompt engineering to improve response relevance and accuracy.
- Integrated the model into a **Flask-based web application** for real-time interaction.
- Collaborated with team members on model evaluation, testing, and optimization.

Tech Stack:
Python, Hugging Face Transformers, LangChain, NLP, LLMs, RAG, Flask, Pandas, NumPy

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2. Book Recommendation System

Team Size: Individual Project
Project Duration: ~4 weeks

Project Summary:
Developed a **Book Recommendation System** using collaborative filtering and rating-based recommendation techniques. The system suggests books based on user preferences and past ratings.

Key Work:

- Cleaned and analyzed book rating datasets.
- Implemented collaborative filtering using **Scikit-Learn**.
- Built a **Flask-based UI** to allow users to receive personalized book recommendations.

Tech Stack:
Python, Pandas, NumPy, Scikit-Learn, Flask

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3. Human vs AI Text Classification System

Team Size: Individual Project
Project Duration: ~3 weeks

Project Summary:
Built an ML-based classification system to differentiate between **human-written and AI-generated text**.

Key Work:

- Preprocessed and labeled text datasets.
- Trained a **Naïve Bayes classification model**.
- Evaluated model performance using accuracy and confusion matrix.
- Deployed the model using Flask for live text classification.

Tech Stack:
Python, NLP, Naïve Bayes, Scikit-Learn, Flask

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4. Recommender Chatbot (GenAI + ML Hybrid Project)

Team Size: Individual Project
Project Duration: ~5 weeks

Project Summary:
Developed a **GenAI-powered chatbot** that combines LLMs with collaborative filtering to provide personalized book recommendations through conversational interaction.

Key Work:

- Integrated LLM-based responses with recommendation logic.
- Fine-tuned prompts to generate context-aware and relevant suggestions.
- Achieved ~90% recommendation relevance based on testing feedback.
- Deployed the chatbot using Flask for real-time usage.

Tech Stack:
Python, LLMs, NLP, Collaborative Filtering, Flask, LangChain

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5. Inventory Demand Forecasting System (Industry Project – Goanny Technologies)

Team Size: Team Project
Project Duration: ~6 months

Project Summary:
Built a **machine learning–based inventory demand prediction system** to optimize inventory planning.

Key Work:

- Performed data cleaning, feature engineering, and EDA.
- Trained regression-based ML models achieving **95% prediction accuracy**.
- Automated inventory planning, reducing stockouts and overstock by **40%**.

Tech Stack:
Python, Pandas, Scikit-Learn, Regression Models, SQL

Additional AI/ML Exposure (Hands-on Work):

- Worked on **text preprocessing pipelines** for NLP tasks
- Practiced **model deployment using Flask**
- Hands-on experience with **EDA, feature engineering, and model evaluation**
- Experience with **Generative AI, prompt engineering, and LLM fine-tuning**
